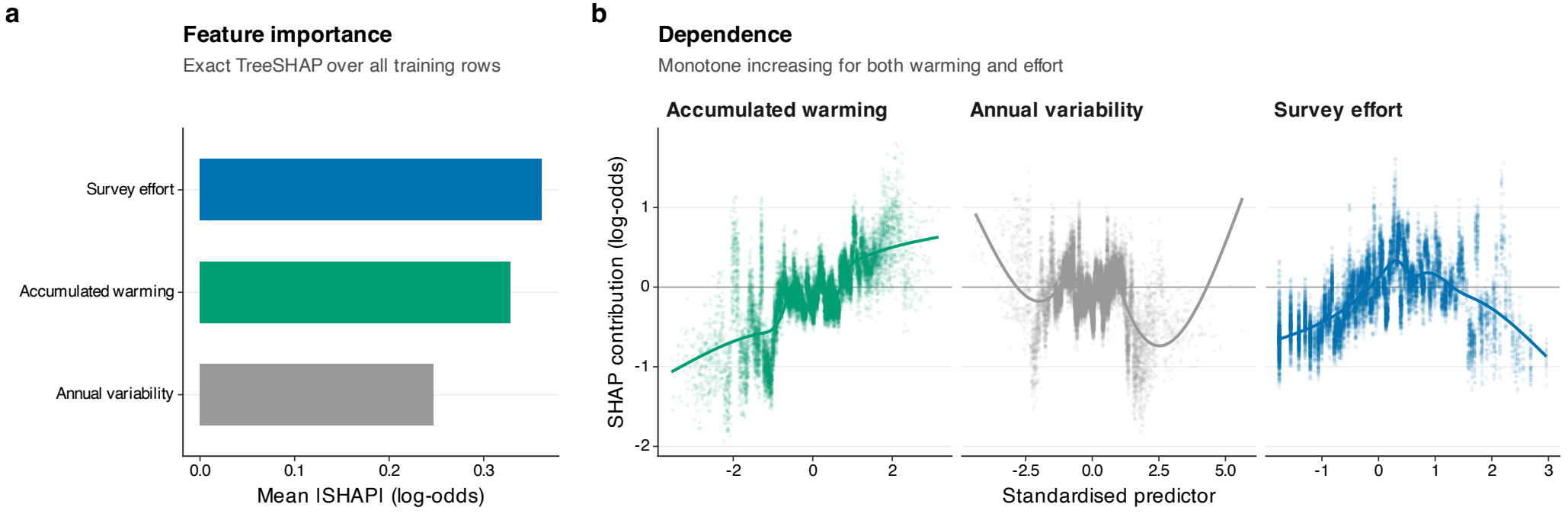

a
Feature importance
Exact TreeSHAP over all training rows
Survey effort
Accumulated warming
Annual variability
0.0
0.1
0.2
0.3
Mean ISHAPI (log-odds)
b
Dependence
Monotone increasing for both warming and effort
Accumulated warming
Annual variability
Survey effort
SHAP contribution (log-odds)
1
0
-1
-2
-2
0
2
-2.5
0.0
2.5
5.0
-1
0
1
2
3
Standardised predictor